

AESS Sustainability Hackathon 2026

Sustainable Electronics for Space Systems

SunSync Power Management Subsystem

1. Project Summary

Field	Aerospace Engineering / Embedded Systems
Project title	SunSync Power Management Subsystem
Team name	LU1-RC
Problem area	Unnecessary battery drain caused by high-power radio transmissions during orbital eclipse phases.
Proposed solution	An intelligent scheduling subsystem that buffers scientific telemetry and transmits it only during optimal orbital windows (Sunlight + Ground Station visibility + Battery Health).
Core proof	Baseline vs. SunSync Battery State of Charge (SoC) comparison over a 24-hour simulation under stress conditions (2W Solar / 4W RF).
Tools used	MATLAB (R2018a), Orbit Visualization Engine.

2. Problem Statement

Scientific CubeSats in Low Earth Orbit (LEO) cycle between sunlight and eclipse every ~95 minutes. Traditional telemetry systems often transmit data on fixed timers or immediately whenever a Ground Station (GS) is visible, regardless of spacecraft power status.

If a high-power transmission (e.g., 4W LoRa/UHF) occurs during the eclipse phase (shadow), the battery must supply 100% of the energy with zero solar input. Over time, these "Eclipse Traps" cause deep battery discharge cycles, leading to thermal stress, reduced mission lifetime, and potential system failure.

3. Proposed Solution

The proposed SunSync Subsystem is a "Store-and-Forward" system located on the main MCU.

- 3.1. Sensing: Continuously monitors Solar Panel Voltage, Battery SoC, and Ground Station visibility.
- 3.2. Buffering: If transmission conditions are poor (Eclipse or Low Battery), data is stored in non-volatile FRAM memory instead of being deleted.
- 3.3. Adaptive Transmission: The RF transmitter activates only when three conditions are met:
 - Solar Energy is available (Solar_Voltage > Threshold).
 - Battery Health is good (SoC > 30%).
 - Ground Station is visible.
- 3.4. Emergency: Critical housekeeping telemetry bypasses the scheduler.

4. System Architecture

Block	Function	Example implementation
Power Management Unit (PMU)	Monitors solar input, battery charge status, and determines orbit phase (Sunlight vs. Eclipse).	PIR sensor or camera-free motion detector
Scientific Payload Sensors	Continuously collects low-priority environmental data for scientific analysis.	LDR / light sensor
Spacecraft Health System	Monitors high-priority spacecraft status for critical alerts.	Arduino Nano / ESP32 / simulation logic
SunSync Telemetry Controller	Buffers data in non-volatile memory (FRAM) and decides when to transmit based on logic.	Relay module or MOSFET switching circuit
RF Transmitter	Outputs telemetry data to Earth in high-power bursts or remains in low-power idle.	LED indicator or simple dashboard

Architecture flow: Scientific Sensors + Health System -> Telemetry Controller (Buffer & Check Sun/GS/SoC) -> RF Transmitter -> Ground Station

5. Operating Logic

- 5.1. The system continuously samples sensors and queues data packets.
- 5.2. Every time step, scheduler evaluates `is_sun` (Orbital Phase) and `is_pass` (GS Visibility).
- 5.3. Baseline Logic: Transmit whenever Ground Station is visible (Day or Night).
- 5.4. SunSync Logic: Transmit only if (`is_sun == TRUE`) AND (`SoC > 30%`) AND (`is_pass == TRUE`).
- 5.5. If conditions fail, queue size increases (buffering).
- 5.6. When conditions are met, the transmitter clears the queue at a high data rate.

6. Baseline vs Optimized Energy Comparison

To demonstrate the necessity of intelligent scheduling, the simulation compares a traditional Baseline system against the optimized SunSync system over a 24-hour operational period.

Device / Mode	Baseline Result	Optimized Result	Comparison
Battery Health (SoC)	Avg: 99.9% (Min: 97.3%)	Avg: 100.0% (Min: 98.6%)	SunSync maintains higher stability, avoiding the gradual voltage dip seen in the baseline (0.1% improvement).
RF Active Time	32 minutes	16 minutes	SunSync successfully reduced transmission time by 50% by avoiding unnecessary passes.
RF Energy Consumption	2.13 Wh	1.07 Wh	Significant energy saving of 50% on the RF subsystem alone.
Total Energy Consumed(24h)	2.33 Wh	1.25 Wh	Overall system energy usage reduced by 46.4%.

Example result

The simulation output demonstrates that the SunSync system achieves a 46.4% reduction in total energy consumption over 24 hours (from 2.33 Wh to 1.25 Wh). Specifically, while the Baseline system transmitted for 32 minutes (expending 2.13 Wh on RF), the SunSync system only transmitted for 16 minutes (expending 1.07 Wh). The SunSync system maintained a slightly higher State of Charge (98.6% minimum vs. 97.3% minimum), proving that intelligent scheduling not only saves power but also maintains a healthier battery profile by avoiding deep discharge cycles during non-optimal orbital windows. Note: The results are based on the logged simulation output provided. The team should ensure that the specific simulation parameters (Solar Input, RF Power, and GS Pass duration) used to generate these logs are clearly documented in the repository.

7. Implementation Evidence

- 7.1. Visual Simulation: 3D animation of Satellite orbiting Earth, visualizing Eclipse/Link/Status in real-time.
- 7.2. Simulation 1 (Orbit Energy): Proves the environment provides intermittent power (0W during eclipse).
- 7.3. Simulation 2 (Battery SoC): Visualizes the "Slow Leak" of the Baseline system vs. the stability of SunSync.
- 7.4. Simulation 3 (RF Duty Cycle): Shows the specific moment (Pass 2) where Baseline wastes energy and SunSync stays idle.
- 7.5. Simulation 4 (Power Consumption): Quantifies the total energy saved (Watt-hours) over 24 hours.

8. Limitations and Future Improvement

- 8.1. Limitations: Simulation assumes simplified circular orbit; ignores atmospheric drag; assumes perfect line-of-sight; uses constant transmission power (no rate adaptation).
- 8.2. Future Work: Integrate real-time prediction of future eclipse passes; implement variable data rates (Adaptive Coding) based on signal strength.

9. Impact Statement

- 9.1. SunSync proves that timing is as important as hardware. By shifting transmission windows by a few minutes (from Eclipse to Sunlight), we can:
- 9.2. Reduce Battery Stress: Avoid deep discharge cycles.
- 9.3. Extend Mission Lifetime: Keep battery in high-health state for months/years longer.
- 9.4. Enable More Science: Saving energy allows more power for sensor payloads.